

305 Final Project Part 1

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1 Introduction

1.1 Simulation Conditions

Two operating regions were defined for all transfer characteristic simulations:

- **Linear region:** $V_{DS} = 0.05$ V
- **Saturation region:** $V_{DS} = 1.0$ V

1.2 Baseline vs. Final Device Parameters

Table summarizes the baseline and final simulation conditions used in this study. The oxide thickness (t_{ox}) and doping concentration were held constant, while the gate work function and interface charge density were modified.

Table 1: Baseline and final simulation conditions.

Parameter	Baseline	Final
Dielectric thickness (t_{ox})	2 nm	2 nm
Doping concentration (N)	1×10^{13}	1×10^{13}
Work function (WF)	4.2 eV	4.5 eV
Interface charge density ($N_{\text{interface}}$)	-1×10^{11}	-1×10^{12}

1.3 Deliverables

The following deliverables were produced for the baseline and final conditions:

- I_D - V_G curves at $V_{DS} = 0.05$ V (linear region)
- I_D - V_G curves at $V_{DS} = 1.0$ V (saturation region)
- Extracted parameters: V_T , DIBL, I_{ON} , I_{OFF} , and SS

2 Extracted Electrical Parameters

This section summarizes the measured (extracted) device parameters obtained from the simulated I_D - V_G transfer characteristics for both the baseline and final conditions, evaluated in the linear region ($V_{DS} = 0.05$ V) and saturation region ($V_{DS} = 1.0$ V).

2.1 Baseline Condition

Linear Region ($V_{DS} = 0.05$ V).

- Threshold voltage: $V_T = 0.2817$ V
- On current: $I_{ON} = 8.41378 \times 10^{-3}$ mA
- Off current: $I_{OFF} = 8.91363 \times 10^{-7}$ mA
- Subthreshold slope: $SS = 95.783$ mV/dec

Saturation Region ($V_{DS} = 1.0$ V).

- Threshold voltage: $V_T = 0.27195$ V
- On current: $I_{ON} = 3.41876 \times 10^{-2}$ mA
- Off current: $I_{OFF} = 2.60792 \times 10^{-5}$ mA
- Subthreshold slope: $SS = 136.643$ mV/dec

2.2 Final Condition

Linear Region ($V_{DS} = 0.05$ V).

- Threshold voltage: $V_T = 0.5951$ V
- On current: $I_{ON} = 5.39287 \times 10^{-3}$ mA
- Off current: $I_{OFF} = 4.0207 \times 10^{-10}$ mA
- Subthreshold slope: $SS = 88.942$ mV/dec

Saturation Region ($V_{DS} = 1.0$ V).

- Threshold voltage: $V_T = 0.474299$ V
- On current: $I_{ON} = 1.83091 \times 10^{-2}$ mA
- Off current: $I_{OFF} = 2.89677 \times 10^{-8}$ mA
- Subthreshold slope: $SS = 106.584$ mV/dec

2.3 Drain-Induced Barrier Lowering (DIBL)

DIBL was extracted by comparing the threshold voltage shift between the linear and saturation I_D - V_G curves for each device condition:

- Baseline: $DIBL = 129.86$ mV/V
- Final: $DIBL = 145.11$ mV/V

Table 2: Magnitude of change from baseline to final condition (Final – Baseline).

Parameter	Region	Baseline	Final	Δ (Final–Baseline)
V_T	Linear ($V_{DS} = 0.05$ V)	0.2817 V	0.5951 V	0.3134 V
V_T	Sat. ($V_{DS} = 1.0$ V)	0.27195 V	0.474299 V	0.202349 V
I_{ON}	Linear ($V_{DS} = 0.05$ V)	8.41378×10^{-3} mA	5.39287×10^{-3} mA	-3.02091×10^{-3} mA
I_{ON}	Sat. ($V_{DS} = 1.0$ V)	3.41876×10^{-2} mA	1.83091×10^{-2} mA	-1.58785×10^{-2} mA
I_{OFF}	Linear ($V_{DS} = 0.05$ V)	8.91363×10^{-7} mA	4.0207×10^{-10} mA	$-8.9096093 \times 10^{-7}$ mA
I_{OFF}	Sat. ($V_{DS} = 1.0$ V)	2.60792×10^{-5} mA	2.89677×10^{-8} mA	$-2.60502323 \times 10^{-5}$ mA
SS	Linear ($V_{DS} = 0.05$ V)	95.783 mV/dec	88.942 mV/dec	-6.841 mV/dec
SS	Sat. ($V_{DS} = 1.0$ V)	136.643 mV/dec	106.584 mV/dec	-30.059 mV/dec
DIBL	–	129.86 mV/V	145.11 mV/V	+15.25 mV/V

3 Discussion of Changes

3.1 Threshold Voltage (V_T) Shift: Baseline vs. Final

A significant increase in V_T is observed:

$$\Delta V_T^{lin} = 0.5951 - 0.2817 = 0.3134 \text{ V}, \quad \Delta V_T^{sat} = 0.474299 - 0.27195 = 0.202349 \text{ V}.$$

$$V_T = V_{FB} + 2\phi_F + \frac{Q_{dep}}{C_{ox}}, \quad V_{FB} = \Phi_{MS} - \frac{Q_{ox}}{C_{ox}}$$

With unchanged doping and dielectric stack, the dominant shift is

$$\Delta V_T \approx \Delta \Phi_{MS} - \frac{\Delta Q_{ox}}{C_{ox}}.$$

$$\Delta \Phi_{MS} \approx (4.5 - 4.2) \text{ eV} = 0.3 \text{ V}$$

$$\Delta N_{\text{interface}} = (-10^{12}) - (-10^{11}) = -9 \times 10^{11} \text{ cm}^{-2} \Rightarrow \Delta Q_{ox} = q \Delta N_{\text{interface}} \approx -1.44 \times 10^{-7} \text{ C/cm}^2$$

For the measured stack $t_{\text{SiO}_2} = 0.96 \text{ nm}$ ($\epsilon_r = 3.9$) and $t_{\text{HfO}_2} = 2 \text{ nm}$ ($\epsilon_r = 25$):

$$\frac{1}{C_{ox}} = \frac{t_{\text{SiO}_2}}{\epsilon_0 \epsilon_r \text{SiO}_2} + \frac{t_{\text{HfO}_2}}{\epsilon_0 \epsilon_r \text{HfO}_2} \Rightarrow C_{ox} \approx 2.71 \times 10^{-6} \text{ F/cm}^2$$

$$-\frac{\Delta Q_{ox}}{C_{ox}} \approx + \frac{1.44 \times 10^{-7}}{2.71 \times 10^{-6}} \approx 0.053 \text{ V}$$

$$\Delta V_T \approx 0.30 + 0.053 = 0.353 \text{ V}$$

This predicted positive shift is consistent with the measured increase in V_T (the smaller apparent shift in saturation is consistent with DIBL).

3.2 On Current (I_{ON}) Change: Baseline vs. Final

A decrease in I_{ON} is observed:

$$\Delta I_{ON}^{lin} = (5.39287 - 8.41378) \times 10^{-3} = -3.02091 \times 10^{-3} \text{ mA}$$

$$\Delta I_{ON}^{sat} = (1.83091 - 3.41876) \times 10^{-2} = -1.58785 \times 10^{-2} \text{ mA}$$

For small V_{DS} (linear region),

$$I_D \approx \mu C_{ox} \frac{W}{L} \left[(V_G - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right]$$

and in saturation (first-order),

$$I_{D,sat} \approx \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_G - V_T)^2.$$

With geometry and dielectric stack unchanged, $\mu C_{ox} \frac{W}{L}$ is approximately constant, so the dominant change comes from the overdrive

$$V_{OV} = V_G - V_T.$$

Since V_T increases from baseline to final, V_{OV} decreases at the same V_G , giving

$$I_{ON}^{lin} \propto (V_G - V_T), \quad I_{ON}^{sat} \propto (V_G - V_T)^2,$$

which explains the reduction of I_{ON} in both regions. The larger fractional drop in saturation is consistent with the quadratic dependence on $(V_G - V_T)$.

3.3 Off Current (I_{OFF}) Change: Baseline vs. Final

A strong reduction in I_{OFF} is observed:

$$\Delta I_{OFF}^{lin} = (4.0207 \times 10^{-10} - 8.91363 \times 10^{-7}) \approx -8.91 \times 10^{-7} \text{ mA}$$

$$\Delta I_{OFF}^{sat} = (2.89677 \times 10^{-8} - 2.60792 \times 10^{-5}) \approx -2.61 \times 10^{-5} \text{ mA}$$

In the subthreshold region, the drain current is approximately

$$I_D \approx I_0 \exp\left(\frac{V_G - V_T}{nV_T^{(th)}}\right),$$

where $V_T^{(th)} = kT/q \approx 25.9 \text{ mV}$ at 300 K, and n is the subthreshold factor. At $V_G = 0$ (off-state),

$$I_{OFF} \propto \exp\left(\frac{-V_T}{nV_T^{(th)}}\right).$$

Therefore, an increase in threshold voltage produces an exponential decrease in I_{OFF} :

$$\frac{I_{OFF,final}}{I_{OFF,base}} \approx \exp\left(\frac{-\Delta V_T}{nV_T^{(th)}}\right).$$

Using the measured linear-region threshold shift $\Delta V_T^{lin} = 0.3134 \text{ V}$ and estimating

$$n \approx \frac{SS}{60 \text{ mV/dec}} \approx \frac{95.783}{60} \approx 1.60,$$

$$\frac{I_{OFF,final}}{I_{OFF,base}} \approx \exp\left(\frac{-0.3134}{(1.60)(0.0259)}\right) \approx 5.2 \times 10^{-4}.$$

The measured ratio is

$$\frac{I_{OFF,final}}{I_{OFF,base}} = \frac{4.0207 \times 10^{-10}}{8.91363 \times 10^{-7}} \approx 4.5 \times 10^{-4},$$

which is consistent with the exponential dependence of subthreshold current on V_T .

3.4 Subthreshold Slope (SS) Change: Baseline vs. Final

Only a modest change in SS is observed:

$$\Delta SS^{lin} = 88.942 - 95.783 = -6.841 \text{ mV/dec}, \quad \Delta SS^{sat} = 106.584 - 136.643 = -30.059 \text{ mV/dec}.$$

The ideal subthreshold slope is

$$SS = \ln(10) \frac{kT}{q} \left(1 + \frac{C_{dep} + C_{it}}{C_{ox}} \right),$$

with $\ln(10) \frac{kT}{q} \approx 60 \text{ mV/dec}$ at 300 K. Since the dielectric stack is unchanged (C_{ox} approximately constant) and the substrate doping is unchanged (C_{dep} approximately constant), SS is expected to stay roughly the same between baseline and final conditions. The work function and fixed interface charge mainly shift V_T through V_{FB} and do not directly change the SS expression.

The measured differences are most likely due to short-channel/leakage effects and the SS fitting range used by the simulator (especially at $V_{DS} = 1 \text{ V}$), rather than a change in C_{ox} or doping.

3.5 Drain-Induced Barrier Lowering (DIBL) Change: Baseline vs. Final

A small increase in DIBL is observed:

$$\Delta \text{DIBL} = 145.11 - 129.86 = 15.25 \text{ mV/V}.$$

DIBL is defined from the threshold shift between the linear and saturation drain biases:

$$\text{DIBL} = \frac{V_{T,lin} - V_{T,sat}}{V_{DS,sat} - V_{DS,lin}}.$$

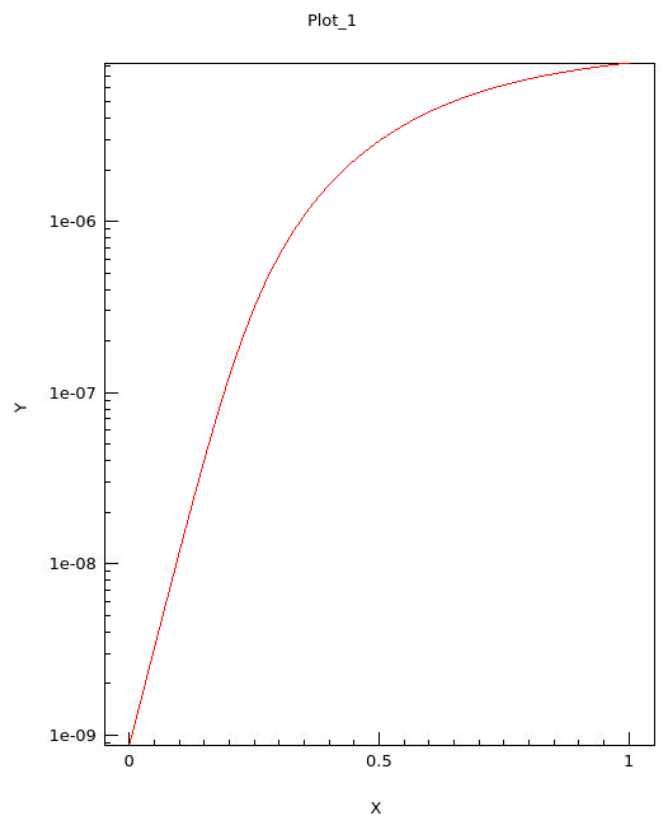
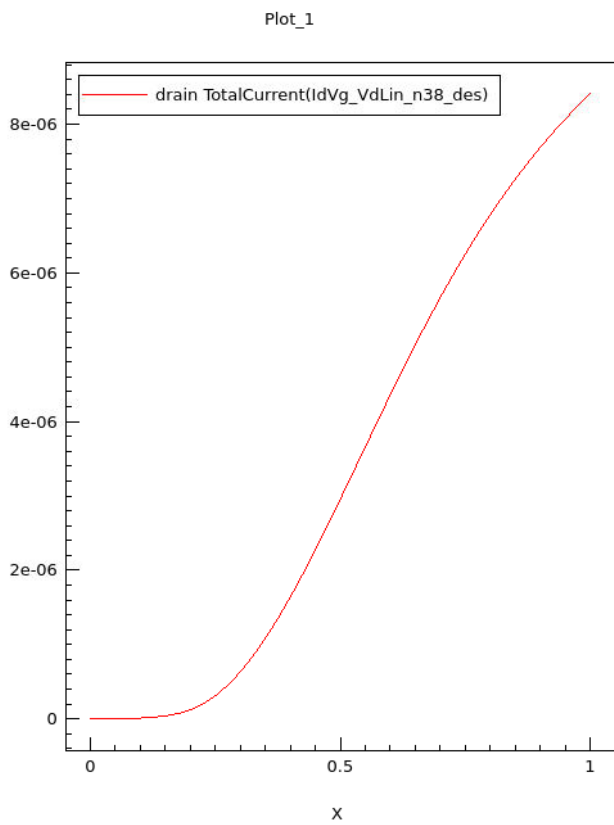
Using the extracted values,

$$\text{DIBL}_{base} = \frac{0.2817 - 0.27195}{1.0 - 0.05} \approx 10.3 \text{ mV/V}, \quad \text{DIBL}_{final} = \frac{0.5951 - 0.474299}{1.0 - 0.05} \approx 127.2 \text{ mV/V}.$$

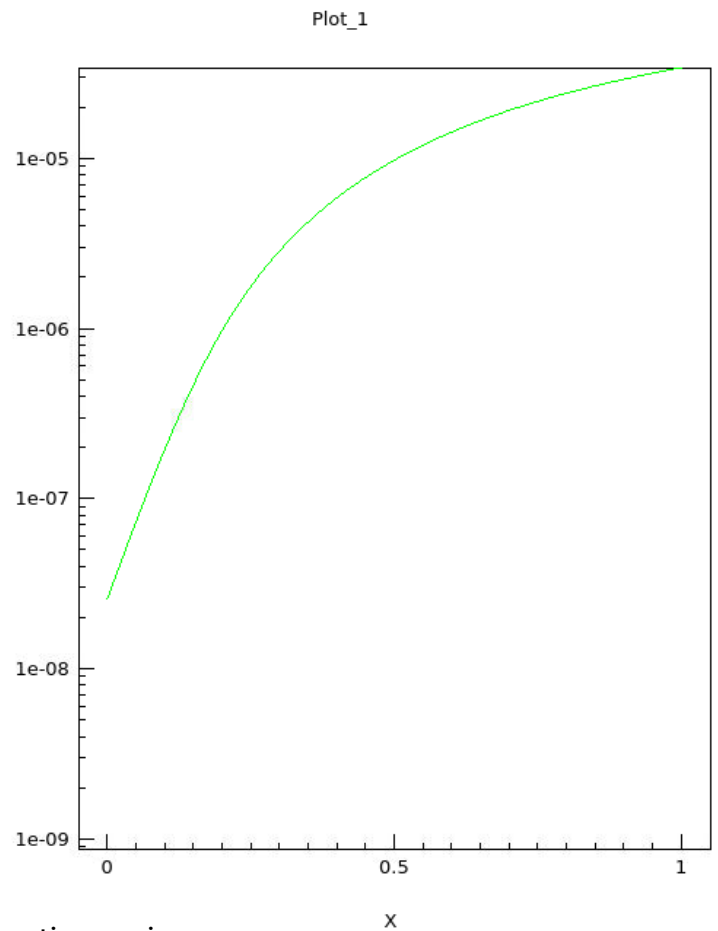
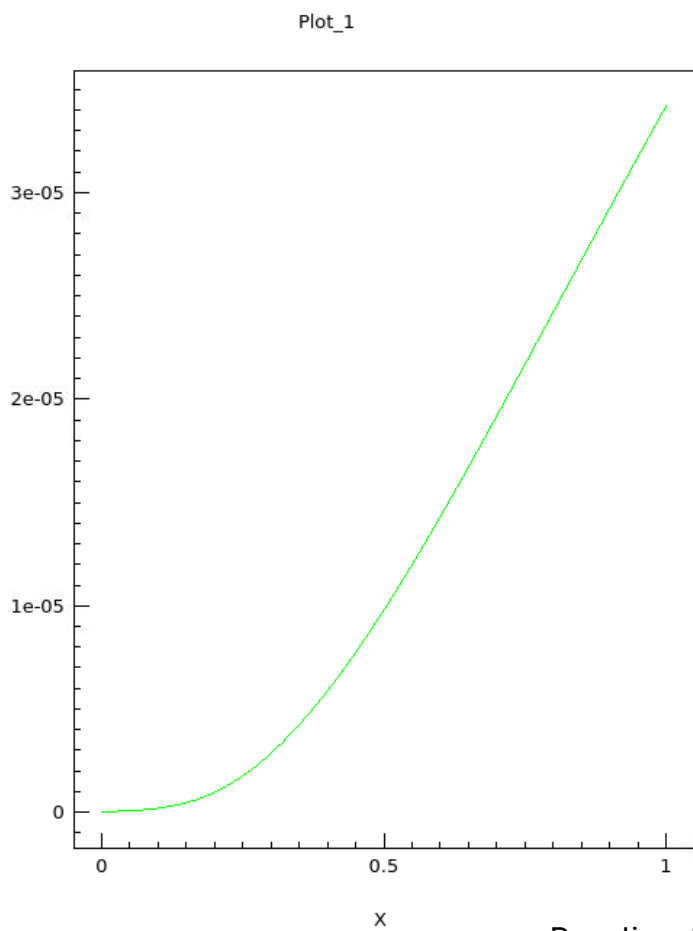
A common short-channel model is

$$V_T(V_{DS}) \approx V_{T0} - \eta V_{DS},$$

so $\text{DIBL} \approx \eta$. With geometry and the dielectric stack unchanged, DIBL is expected to be mainly set by device electrostatics/short-channel geometry, not by a flat-band shift. The reported DIBL values (129.86 mV/V and 145.11 mV/V) show only a modest change; the mismatch with the simple hand calculation above can come from human error in measuring the VT values.

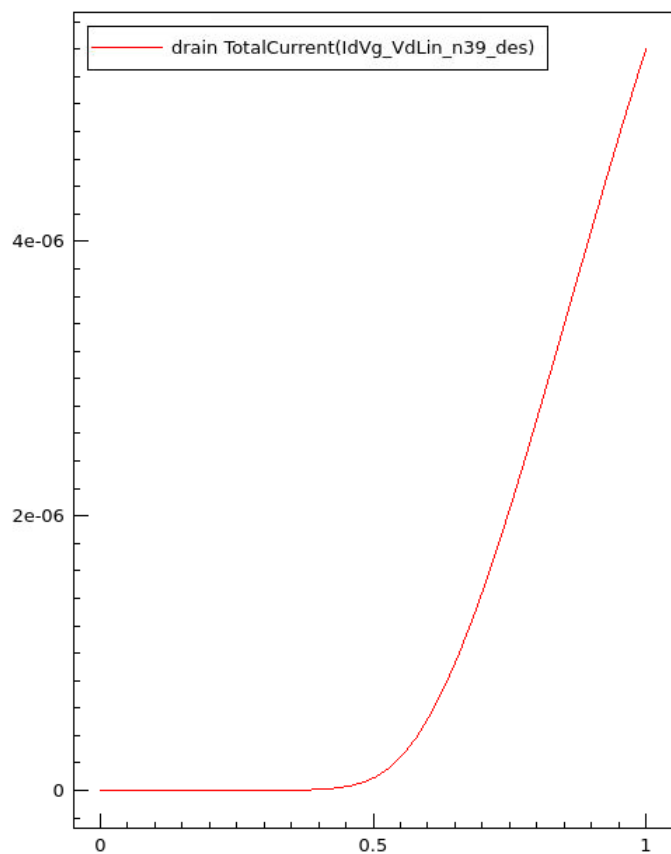


Baseline, Linear region

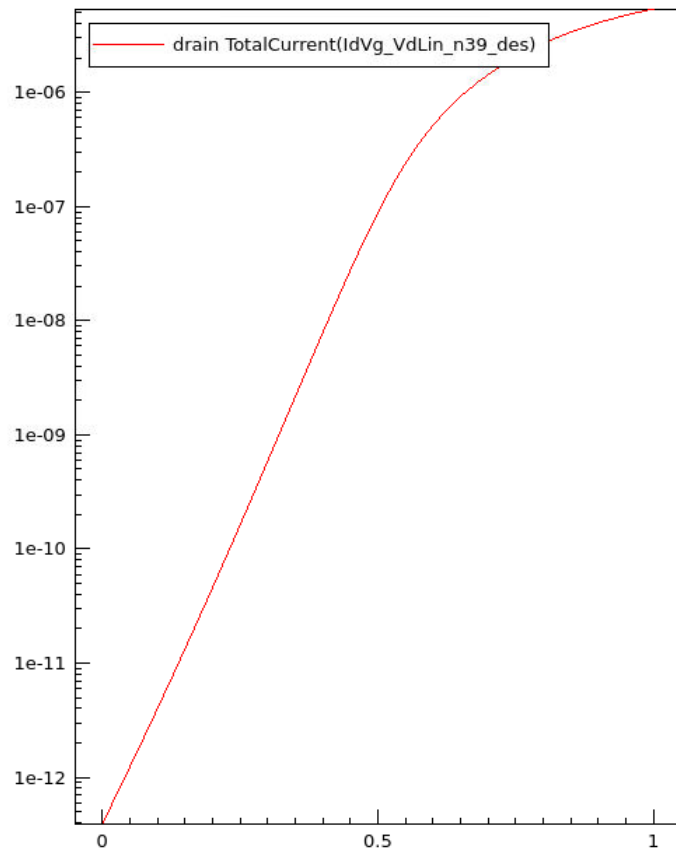


Baseline, Saturation region

Plot_1

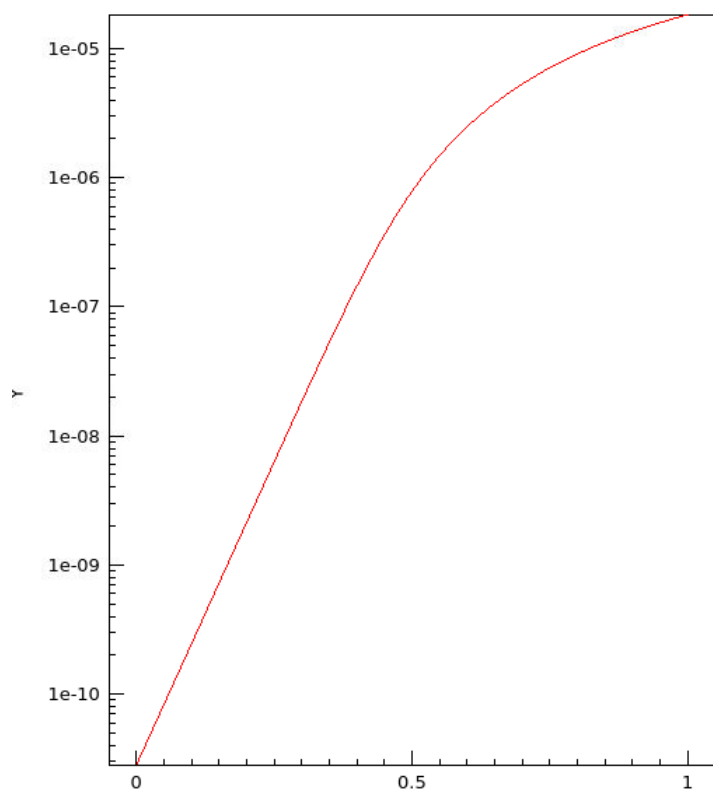
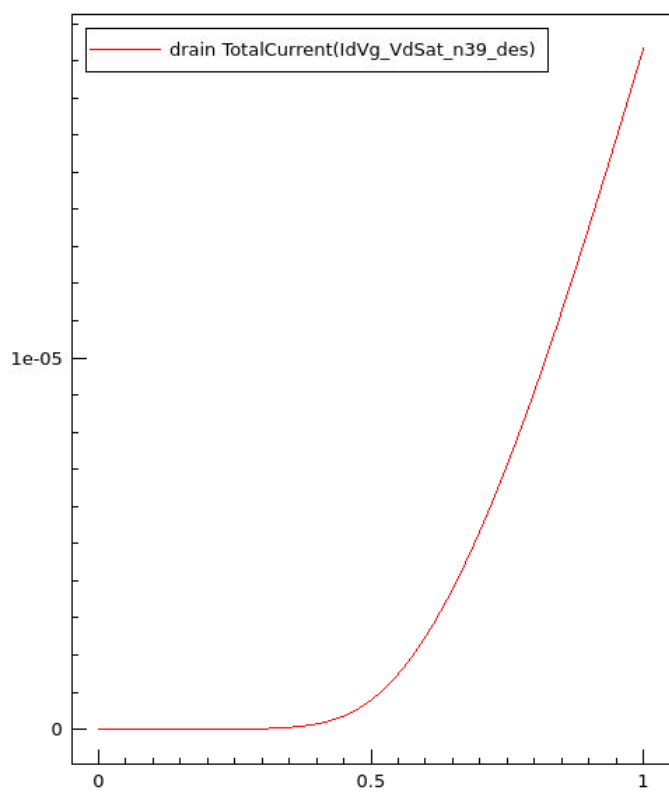


Plot_1



Final, Linear region

Plot_1



Final, Saturation region

305 Final Project Part 2

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1 Introduction

Part II of this project focuses on the effect of *interface traps* (energy-distributed acceptor traps at the Oxide/Silicon interface) on the FinFET transfer characteristics and extracted parameters. Unlike Part I, where the interface traps behaved as fixed charge (Q_{ox}), Part II introduces energy-dependent traps described by a concentration N_{trap} (units: $\text{cm}^{-2}\text{eV}^{-1}$) centered at midgap ($E_0 = 0$) with an energy spread E_s . The **baseline condition** for Part II uses the same device settings as Part I, with the default trap distribution parameters $E_0 = 0$ and $E_s = 0.15$. From Part I (baseline, linear region at $V_{DS} = 0.05 \text{ V}$), the extracted values were:

$$V_{T,base}^{lin} = 0.2817 \text{ V}, \quad SS_{base}^{lin} = 95.783 \text{ mV/dec.}$$

In saturation ($V_{DS} = 1 \text{ V}$), the baseline values were:

$$V_{T,base}^{sat} = 0.27195 \text{ V}, \quad SS_{base}^{sat} = 136.643 \text{ mV/dec.}$$

The **final condition** for Part II modifies the interface-trap parameters to

$$N_{\text{trap}} = 1.8 \times 10^{13} \text{ cm}^{-2}\text{eV}^{-1}, \quad E_s = 1.1, \quad E_0 = 0,$$

while keeping the remaining device parameters the same as the baseline.

2 Extracted Values (Part II)

This section summarizes the key extracted quantities used for the analytical verification of V_T and SS in Part II.

2.1 Electrical Parameters from Simulation

Linear Region ($V_{DS} = 0.05 \text{ V}$).

$$V_T^{lin} = 0.857243 \text{ V}, \quad SS^{lin} = 165.994 \text{ mV/dec}$$

Saturation Region ($V_{DS} = 1.0 \text{ V}$).

$$V_T^{sat} = 0.836682 \text{ V}, \quad SS^{sat} = 186.774 \text{ mV/dec}$$

2.2 Extracted Structure Parameters

$$N_A = 4.052018 \times 10^{18} \text{ cm}^{-3}$$

$$t_{\text{SiO}_2} = 0.96 \text{ nm}$$

2.3 Subthreshold Slope (SS) Verification

Measured SS:

$$SS_{base}^{lin} = 95.783 \text{ mV/dec}, \quad SS_{final}^{lin} = 165.994 \text{ mV/dec}$$

$$SS_{final}^{sat} = 186.774 \text{ mV/dec}$$

$$SS = S_0 \left(1 + \frac{C_{dep} + C_{it}}{C_{ox}} \right), \quad S_0 = \ln(10) \frac{kT}{q} \approx 59.6 \text{ mV/dec} \quad (300 \text{ K})$$

Using the measured gate stack,

$$C_{ox} \approx 2.71 \times 10^{-6} \text{ F/cm}^2$$

and the extracted doping $N_A = 4.052018 \times 10^{18} \text{ cm}^{-3}$,

$$\phi_F = \frac{kT}{q} \ln\left(\frac{N_A}{n_i}\right), \quad \psi_s \approx 2\phi_F, \quad W_d = \sqrt{\frac{2\varepsilon_{si}\psi_s}{qN_A}}, \quad C_{dep} = \frac{\varepsilon_{si}}{W_d}$$

$$C_{dep} \approx 5.72 \times 10^{-7} \text{ F/cm}^2$$

Baseline SS $\Rightarrow$ intrinsic interface-trap capacitance:

$$C_{it,0} = C_{ox} \left(\frac{SS_{base}^{lin}}{S_0} - 1 \right) - C_{dep} \approx 1.07 \times 10^{-6} \text{ F/cm}^2$$

Final interface traps: $N_{trap} = 1.8 \times 10^{13} \text{ cm}^{-2} \text{ eV}^{-1}$, $E_s = 1.1$

$$D_{it,added} \approx N_{trap} E_s \Rightarrow C_{it,added} \approx q D_{it,added} = q N_{trap} E_s \approx 3.17 \times 10^{-6} \text{ F/cm}^2$$

$$C_{it,total} = C_{it,0} + C_{it,added} \approx (1.07 + 3.17) \times 10^{-6} = 4.25 \times 10^{-6} \text{ F/cm}^2$$

Predicted SS (linear region):

$$SS_{pred}^{lin} = S_0 \left(1 + \frac{C_{dep} + C_{it,total}}{C_{ox}} \right) \approx 59.6 \left(1 + \frac{5.72 \times 10^{-7} + 4.25 \times 10^{-6}}{2.71 \times 10^{-6}} \right) \approx 165.5 \text{ mV/dec}$$

$$SS_{sim}^{lin} = 165.994 \text{ mV/dec}$$

The saturation-region SS is larger in simulation ($SS_{final}^{sat} = 186.774 \text{ mV/dec}$) due to drain-related non-idealities at high V_{DS} ; the linear-region SS provides the closest match to the ideal analytical expression above.

2.4 Threshold Voltage (V_T) Verification

Measured threshold voltages:

$$V_{T,base}^{lin} = 0.2817 \text{ V}, \quad V_{T,final}^{lin} = 0.857243 \text{ V}$$

$$\Delta V_T^{lin} = 0.857243 - 0.2817 = 0.575543 \text{ V}$$

$$V_T = V_{FB} + 2\phi_F + \frac{|Q_{dep}|}{C_{ox}}, \quad V_{FB} = \Phi_{MS} - \frac{Q_{ox}}{C_{ox}}$$

$$\Rightarrow V_T = \Phi_{MS} + 2\phi_F + \frac{|Q_{dep}|}{C_{ox}} - \frac{Q_{ox}}{C_{ox}}$$

Using the extracted stack capacitance from the measured $\text{SiO}_2/\text{HfO}_2$ thicknesses:

$$C_{ox} \approx 2.71 \times 10^{-6} \text{ F/cm}^2$$

Using the extracted doping $N_A = 4.052018 \times 10^{18} \text{ cm}^{-3}$:

$$\phi_F = \frac{kT}{q} \ln\left(\frac{N_A}{n_i}\right) \approx 0.513 \text{ V}, \quad |Q_{dep}| = \sqrt{4\epsilon_{si}qN_A\phi_F} \approx 1.18 \times 10^{-6} \text{ C/cm}^2$$

$$\Phi_{MS} = \Phi_M - \Phi_S, \quad \Phi_S = \chi + \frac{E_g}{2} + \phi_F$$

Using $\chi = 4.05 \text{ eV}$, $E_g = 1.12 \text{ eV}$, $\Phi_M = 4.2 \text{ eV}$:

$$\Phi_{MS} \approx 4.2 - (4.05 + 0.56 + 0.513) = -0.923 \text{ V}$$

Intrinsic (baseline) charge term extracted from $V_{T,base}^{lin}$:

$$Q_{ox,0} = C_{ox} \left(\Phi_{MS} + 2\phi_F + \frac{|Q_{dep}|}{C_{ox}} - V_{T,base}^{lin} \right) \approx 6.91 \times 10^{-7} \text{ C/cm}^2$$

Added interface traps (final): $N_{trap} = 1.8 \times 10^{13} \text{ cm}^{-2} \text{ eV}^{-1}$, $E_s = 1.1$

$$D_{it,added} \approx N_{trap}E_s, \quad Q_{it,added} \approx -q \frac{D_{it,added}}{2} = -q \frac{N_{trap}E_s}{2} \approx -1.59 \times 10^{-6} \text{ C/cm}^2$$

$$\Delta V_{T,added} \approx -\frac{Q_{it,added}}{C_{ox}} \approx \frac{1.59 \times 10^{-6}}{2.71 \times 10^{-6}} \approx 0.584 \text{ V}$$

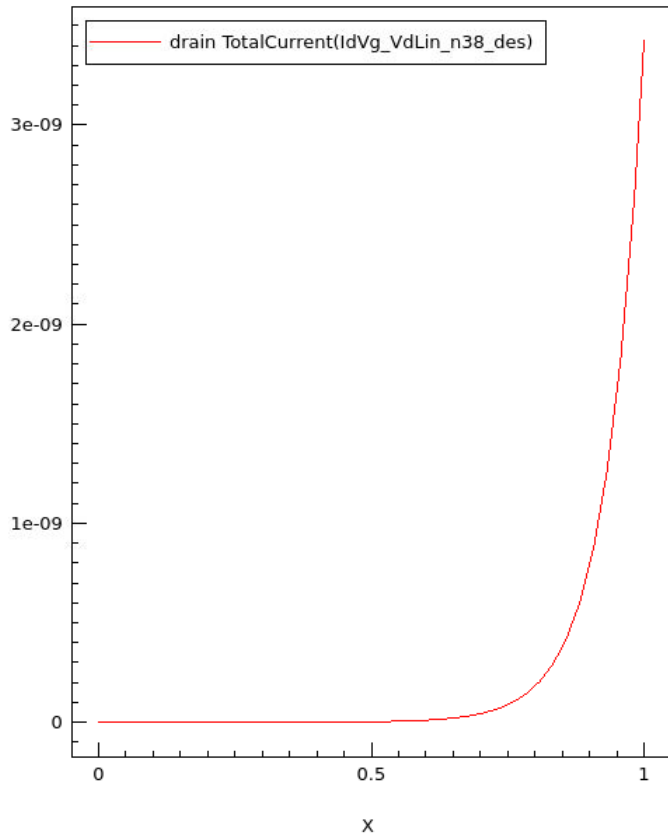
Predicted final threshold voltage:

$$V_{T,pred}^{lin} \approx V_{T,base}^{lin} + \Delta V_{T,added} = 0.2817 + 0.584 = 0.866 \text{ V}$$

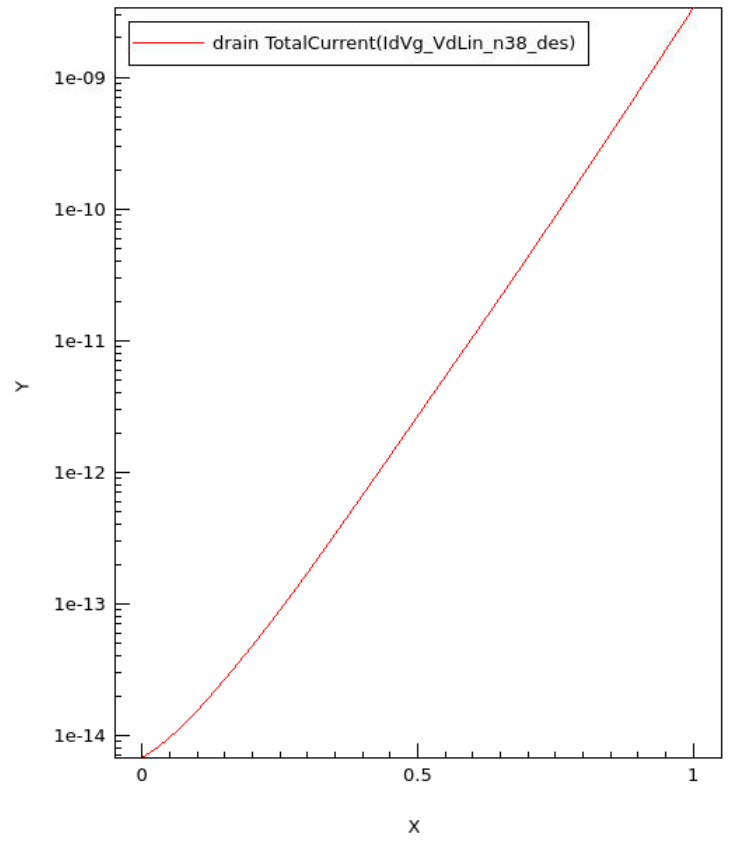
$$V_{T,sim}^{lin} = 0.857243 \text{ V}$$

The linear-region value ($V_{DS} = 0.05 \text{ V}$) is used for the analytical V_T match. The saturation-extracted V_T ($V_{DS} = 1 \text{ V}$) is expected to be slightly smaller due to drain-induced barrier lowering at high V_{DS} .

Plot_1

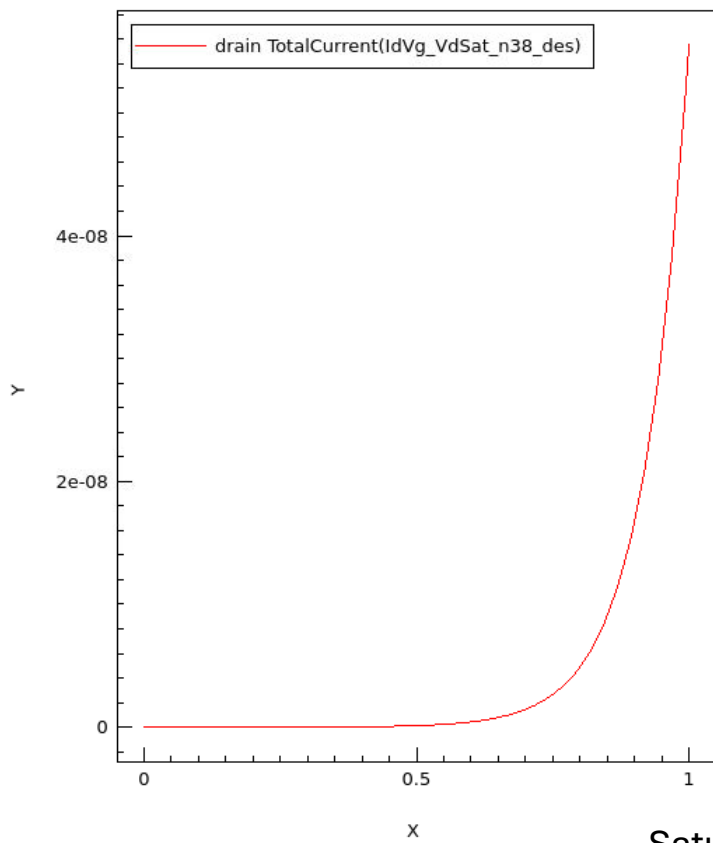


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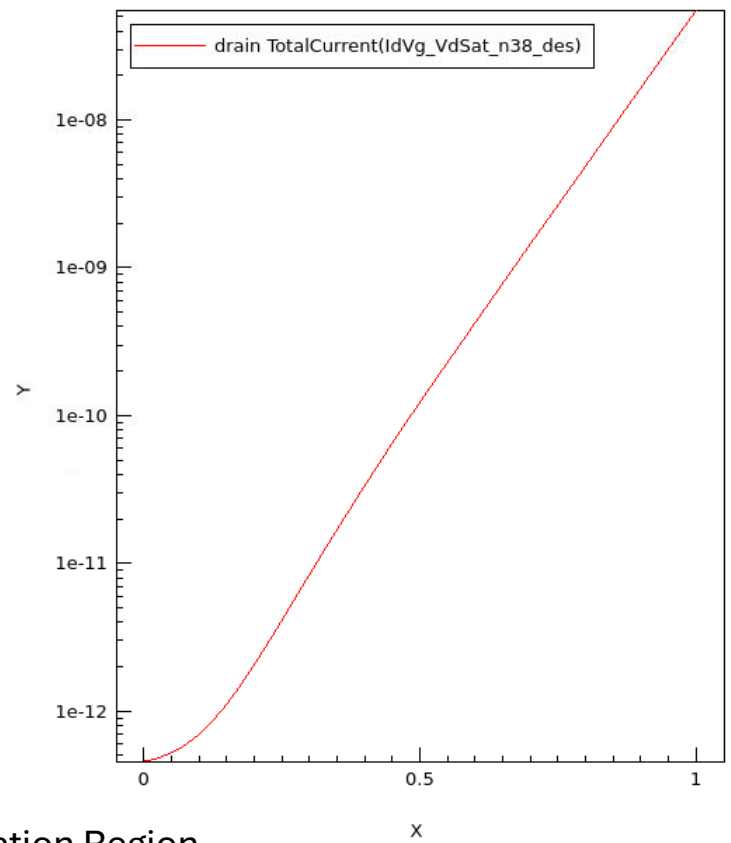


Linear Region

Plot_1



Plot_1



Saturation Region